Nusantara Medical Science Journal

Diabetic_Medicine_Manuscript.pdf

Manuscript 87

Nusantara Medical Science Jounal

Universitas Hasanuddin

Document Details

Submission ID
trn:oid:::1:3673715861

Submission Date
Oct 9, 2026, 10:10 AM GMT+7

Download Date
Oct 9, 2026, 10:28 AM GMT+7

File Name
Diabetic_Medicine_Manuscript.pdf

File Size
285.3 KB

16 Pages

4,436 Words

24,707 Characters

27% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Exclusions

- 8 Excluded Matches

Match Groups

73 Not Cited or Quoted 25%
Matches with neither in-text citation nor quotation marks

9 Missing Quotations 2%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

25% Internet sources
20% Publications
5% Submitted works (Student Papers)

Match Groups

- **73** Not Cited or Quoted 25%
  Matches with neither in-text citation nor quotation marks
- **9** Missing Quotations 2%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

25% Internet sources
20% Publications
5% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	www.mdpi.com 7%
2	Internet	www.frontiersin.org 2%
3	Internet	mdpi-res.com 1%
4	Internet	www.medrxiv.org 1%
5	Publication	Soudabeh Hosseini, Mojtaba Babaalian, Hamideh Sadat Mirmohammadi, Mehdi K... <1%
6	Internet	lirias.kuleuven.be <1%
7	Internet	pmc.ncbi.nlm.nih.gov <1%
8	Internet	www.omjournal.org <1%
9	Internet	www.physiciansweekly.com <1%
10	Internet	www.researchgate.net <1%

11	Internet	
www.researchsquare.com		<1%
12	Internet	
horizonepublishing.com		<1%
13	Internet	
assets-eu.researchsquare.com		<1%
14	Internet	
public-pages-files-2025.frontiersin.org		<1%
15	Publication	
William T. Cefalu, Jack L. Leahy. "Insulin Therapy", CRC Press, 2019		<1%
16	Internet	
www.oncotarget.com		<1%
17	Internet	
www.thieme-connect.com		<1%
18	Publication	
"Biomarkers in Diabetes", Springer Science and Business Media LLC, 2023		<1%
19	Publication	
Aneta Szofer-Sendrowska, Martyna Kaja Sendrowska, Damian Sendrowski, Bartos...		<1%
20	Internet	
www.nature.com		<1%
21	Internet	
www.scopus.com		<1%
22	Publication	
Li Qin, Zhen Yang, Hongxia Gu, Shuai Lu, Qun Shi, Yin Xing, Xiaoyong Li, Rui Li, Gu...		<1%
23	Internet	
eprints.ssu.ac.ir		<1%
24	Internet	
racs-asc.evexus.net		<1%

25 Internet
aacrjournals.org <1%

26 Internet
iranianra.ir <1%

27 Internet
d.docksci.com <1%

28 Internet
escholarship.org <1%

29 Internet
f1000research-files.f1000.com <1%

30 Publication
Atefeh Jannatalipour, Atefeh Amrollahi Bioky, Nafiseh Tavasoli, Farima Farsi et al.... <1%

31 Publication
Hind Abdul Hussain, Ahmed Naseer Kaftan. "The TCF7L2 rs12255372 variant as a ... <1%

32 Internet
vbn.aau.dk <1%

33 Publication
Massimo Volpe. "Cardiovascular Disease Prevention", CRC Press, 2026 <1%

34 Publication
Nelci Ros Maniagasi. "Association of TCF7L2 Rs12255372 with Obesity and Adiposi... <1%

35 Internet
dergipark.org.tr <1%

36 Internet
journal-acmj.com <1%

37 Internet
pubmed.ncbi.nlm.nih.gov <1%

38 Internet
www.aem-sbem.com <1%

39 Internet

www.dovepress.com <1%

40 Internet

www.thejas.com.pk <1%

41 Publication

Barry J. Goldstein, Dirk Mueller-Wieland. "Type 2 Diabetes - Principles and Practic... <1%

42 Publication

Yung-Chuan Lu, Teng-Hung Yu, Chin-Feng Hsuan, Chia-Chang Hsu et al. "Associati... <1%

43 Publication

Erfan Sabouri, Elahe Kaveh, Seyyed Reza Raeeskarami, Vahid Ziaee, Banafsheh Se... <1%

44 Internet

centaur.reading.ac.uk <1%

45 Publication

Liong Boy Kurniawan, Winda Mulyamin, Arif Santoso, Ika Yustisia. "Serum uric aci... <1%

46 Internet

journals.lww.com <1%

Population-specific distribution of transcription factor 7-like 2 rs12255372 in Indonesian adults with and without Type 2 diabetes

Running title: TCF7L2 rs12255372 in Indonesian adults

Prince Saputra[1], Miswar Fattah[2], Himawan Sanusi[3,4], Andi Alfian Zainuddin[5], Rusdina Bte Ladju[6], Mansyur Arif[7] and Liong Boy Kurniawan[7]*

[1] Doctoral Program, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[2] Research and Specialty Test Development Division, PT Prodia Widyahusada Tbk, Jakarta 10430, Indonesia

[3] Department of Internal Medicine, Faculty of Medicine, Hasanuddin University, Makassar, Indonesia

[4] Department of Community Medicine, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[5] Department of Pharmacology, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[6] Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[7] Dr. Wahidin Sudirohusodo National General Hospital, Hasanuddin University, Makassar, Indonesia

Corresponding author: Liong Boy Kurniawan, Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Perintis Kemerdekaan Street Km. 10, Tamalanrea, Makassar 90245, South Sulawesi, Indonesia; Email: liongboykurniawan@yahoo.com

ORCID iDs: Prince Saputra, 0009-0004-2891-7002; Miswar Fattah, 0000-0003-3679-5901; Himawan Sanusi, 0000-0003-3828-2958; Andi Alfian Zainuddin, 0000-0003-4436-3906; Rusdina Bte Ladju, 0000-0001-5393-0429; Mansyur Arif, 0000-0002-9104-8127; Liong Boy Kurniawan, 0000-0002-2633-0441

Main text word count: 2069

Abstract word count: 224

Conflicts of interest

The authors declare that they have no conflicts of interest.

Novelty statement

- **What is already known?** TCF7L2 variants are among the most reproducible common genetic signals for Type 2 diabetes, but the frequency and effect of rs12255372 vary substantially between populations and Indonesian data are scarce.
- **What this study has found?** The rs12255372 T allele was rare in this Indonesian cohort, no TT genotype was observed, and there was no detectable association with Type 2 diabetes, C-peptide or HOMA-IR.
- **What are the implications?** Diabetes-associated variants identified in other populations should be locally validated before use in risk stratification in Indonesia.

Acknowledgements

This study was supported by the Pendidikan GratisPol Generasi Emas Scholarship Program of the Government of East Kalimantan Province, Indonesia. During manuscript preparation, ChatGPT (OpenAI) was used only to assist with language clarity, readability and organisation. The authors reviewed and edited all output and take full responsibility for the final manuscript.

Abstract

Aims: To examine the population-specific distribution of transcription factor 7-like 2 (TCF7L2) rs12255372 and its associations with Type 2 diabetes, C-peptide and insulin resistance in Indonesian adults.

Methods: This case-control study included 181 adults: 111 with Type 2 diabetes, 60 controls without diabetes and 10 participants with Type 1 diabetes included only as a descriptive metabolic comparison group. Genotypes were determined by polymerase chain reaction followed by Sanger sequencing. Fasting glucose, glycated haemoglobin, insulin, C-peptide and homeostasis model assessment of insulin resistance (HOMA-IR) were assessed. The primary genetic analysis compared Type 2 diabetes cases with controls, and multivariable logistic regression adjusted for age, body mass index and sex.

Results: GG was present in 96% of Type 2 diabetes cases and 97% of controls; GT occurred in 3.6% and 3.3%, respectively, and no TT genotype was detected. rs12255372 was not associated with Type 2 diabetes (odds ratio 1.08, 95% CI 0.19, 6.16; P = 0.930). C-peptide and HOMA-IR did not differ significantly between GG and GT carriers (P = 0.157 and P = 0.609), and genotype was not significant in the adjusted model.

Conclusions: The rs12255372 T allele was uncommon in this Indonesian cohort, and no detectable association with Type 2 diabetes or the assessed metabolic phenotypes was identified. Larger multicentre studies are needed to evaluate the transferability of diabetes-associated genetic markers across Indonesian populations.

Keywords: Transcription Factor 7-Like 2 Protein; Diabetes Mellitus, Type 2; C-Peptide; Insulin Resistance; Genetic Association Studies; Indonesia

Introduction

Type 2 diabetes is a chronic metabolic disorder arising from impaired insulin secretion, insulin resistance, environmental exposures and genetic susceptibility. It is a major contributor to premature mortality, disability and healthcare expenditure. The global burden continues to rise, with particularly rapid increases in low- and middle-income countries, including Indonesia.[1-4] Because genetic architecture differs between populations, locally generated evidence is needed before risk variants identified elsewhere are used for prediction or stratification.

Among established Type 2 diabetes susceptibility loci, transcription factor 7-like 2 (TCF7L2) is one of the most consistently replicated. TCF7L2 is located on chromosome 10q25.2-q25.3 and encodes a transcription factor involved in Wnt/beta-catenin signalling, incretin responsiveness, pancreatic beta-cell activity, insulin secretion and glucose homeostasis.[5-8] The intronic single-nucleotide polymorphism rs12255372 (G>T) has been associated with Type 2 diabetes and altered beta-cell function in several populations, but allele frequency and effect estimates vary considerably according to population background, linkage disequilibrium structure and environmental context.[9-15]

Evidence from Indonesia remains limited, particularly studies that combine rs12255372 genotyping with metabolic indicators of endogenous insulin secretion and insulin resistance. C-peptide reflects endogenous insulin production, whereas the homeostasis model assessment of insulin resistance (HOMA-IR) provides an estimate of insulin resistance.[16-18] We therefore examined the distribution of rs12255372 and its association with Type 2 diabetes, C-peptide and HOMA-IR in Indonesian adults. A small Type 1 diabetes group was retained only as a descriptive metabolic comparison group to illustrate severe beta-cell dysfunction.

Methods

Ethics and study design

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Health Research Ethics Committee of Hasanuddin University, Indonesia (approval no. 563/UN4.6.4.5.31/PP36/2025). Written informed consent was obtained from all participants before enrolment.

This observational case-control study was conducted between July and December 2025. Participants were recruited and blood samples were collected in Samarinda, East Kalimantan, Indonesia. Recruitment took place at healthcare facilities and research laboratories specified in the study protocol. Participants were selected according to predefined inclusion and exclusion criteria. Laboratory examinations were performed at the Biomolecular Teaching Hospital Laboratory of HUM-RC, Hasanuddin University, Makassar, South Sulawesi, Indonesia. The study size reflected the total number of eligible participants recruited during July–December 2025 for whom the required study data were available. A total of 181 adults were enrolled: 111 with Type 2 diabetes,

60 controls without diabetes and 10 with Type 1 diabetes included only as a descriptive metabolic comparison group. Participants were men and women aged 18-49 years; 29 were men and 152 were women.

Type 2 diabetes was diagnosed according to the American Diabetes Association Standards of Care in Diabetes-2025.[19] Controls had fasting blood glucose levels consistent with the absence of diabetes and no previous history of diabetes. Participants with Type 1 diabetes were identified by a physician on the basis of the clinical diagnosis documented in the medical record, with age considered as part of the clinical assessment. Eligible participants were men and women aged 18-49 years who provided written informed consent. Individuals with severe systemic illness, pregnancy, malignancy or acute infection were excluded. Participants with insufficient data for a particular analysis were excluded from that analysis.

Clinical and biochemical measurements

Peripheral venous blood was collected after an overnight fast. Fasting blood glucose was measured using a BIOLIS 50i Superior automated clinical analyser (Biolis Medical Corp., Hino, Tokyo, Japan) with an enzymatic method. Glycated haemoglobin (HbA1c) was measured using an ADAMS HA-8380V glycohaemoglobin analyser (ARKRAY Factory, Inc., Koka, Shiga, Japan) based on ion-exchange chromatography. Fasting insulin and C-peptide concentrations were measured using a VIRTUEDX VERCENTRA CS-1500 analyser (PT Darma Labora Indonesia, Cikarang, Bekasi, West Java, Indonesia) with a chemiluminescence immunoassay method. HOMA-IR was calculated as fasting insulin (mIU/L) multiplied by fasting glucose (mmol/L), divided by 22.5. C-peptide was used as an indicator of endogenous insulin secretion.

DNA extraction, genotyping and sequence analysis

Peripheral venous blood (3-5 mL) was collected in EDTA tubes. Genomic DNA was extracted from 200 µL of whole blood using the Geneaid Genomic DNA Mini Kit (Blood/Cultured Cell; Cat. No. GB100; Geneaid Biotech Ltd., New Taipei City, Taiwan) according to the manufacturer's instructions. The procedure included proteinase K digestion, cell lysis, binding to a silica membrane, washing and elution in 100 µL of elution buffer. Purified DNA was stored at -40 °C until amplification.

A 346-bp fragment containing TCF7L2 rs12255372 was amplified using the forward primer 5'-CTGGAAACTAAGGCGTGAGG-3' and reverse primer 5'-GGGTCGATGTTGTTGAGCTT-3'. PCR amplification was performed using a 2x PCR Master Mix containing Taq DNA polymerase in a T100 Thermal Cycler (Bio-Rad Laboratories, Inc., Hercules, CA, USA). PCR comprised initial denaturation at 95 °C for 5 min; 40 cycles of denaturation at 95 °C for 15 s, annealing at 65 °C for 15 s and extension at 72 °C for 45 s; and final extension at 72 °C for 15 min. All 181 PCR-amplified samples underwent Sanger sequencing through 1st BASE Sanger Sequencing Services using an ABI PRISM 3730xl Genetic Analyzer (Applied Biosystems, Foster City, CA, USA).

Chromatograms were analysed using Geneious Prime version 2025.2, and variant identity was verified against NCBI dbSNP rs12255372. The locus was mapped to the GRCh38.p14 human genome assembly and is described in HGVS genomic nomenclature as NC_000010.11:g.113049143G>T. A single guanine peak at the polymorphic locus was classified as GG, whereas overlapping guanine and thymine peaks were classified as GT. No TT genotype was detected.

Statistical analysis

For each analysis, participants were included only if they met the study eligibility criteria and had the information required for the variables analysed. The enrolled sample comprised 181 participants; the number included in an individual analysis depended on the availability of the relevant data or genotype. Continuous variables are presented as mean ± standard deviation or median (interquartile range), according to data distribution, and categorical variables as counts. For comparisons among the Type 2 diabetes, control and Type 1 diabetes groups, normally distributed continuous variables were analysed using one-way analysis of variance and non-normally distributed continuous variables using the Kruskal-Wallis test. Sex distribution was compared using Fisher's exact test.

In the primary case-control analysis, genotype and allele distributions between participants with Type 2 diabetes and controls were compared using Fisher's exact test. Hardy-Weinberg equilibrium was evaluated in the control group. Exploratory genotype-phenotype comparisons of C-peptide and HOMA-IR were performed within the genotyped cohort using participants with the data required for each comparison and the Mann-Whitney U test.

Multivariable logistic regression evaluated independent correlates of Type 2 diabetes case status. Diabetes status was coded as 0 for controls and 1 for Type 2 diabetes; rs12255372 genotype as 0 for GG and 1 for GT; and sex as 0 for women and 1 for men. The association between GT carrier status and Type 2 diabetes was adjusted for age, body mass index (BMI) and sex. Odds ratios (ORs) with 95% confidence intervals (CIs) are reported. Statistical significance was defined as $P < 0.05$. Analyses were performed using IBM SPSS Statistics version 26.0 (IBM Corp., Armonk, NY, USA). Because the rs12255372 T allele was rare and no TT homozygotes were identified, power to detect modest genetic effects was limited. Reporting was guided by the Strengthening the Reporting of Genetic Association Studies (STREGA) recommendations.[20]

Results

Participant characteristics

The available clinical and biochemical characteristics of the enrolled cohort of 181 participants are summarised in Table 1. Participants with Type 2 diabetes were older and had higher fasting glucose, HbA1c, insulin and HOMA-IR values than controls. The Type 1 diabetes descriptive group had markedly lower C-peptide concentrations, consistent with severe loss of endogenous insulin secretion.

Genotype distribution and association with Type 2 diabetes

The GG genotype predominated in both the Type 2 diabetes and control groups. GT was detected in 4 of 111 participants with Type 2 diabetes (3.6%) and 2 of 60 controls (3.3%); no TT genotype was observed. The control genotype distribution did not deviate from Hardy-Weinberg equilibrium ($P = 1.000$). Neither genotype nor allele distribution differed significantly between groups. Compared with GG, GT was not associated with Type 2 diabetes (OR 1.08, 95% CI 0.19, 6.16; $P = 0.930$) (Table 2). Representative chromatograms are shown in Figure 1.

C-peptide and HOMA-IR by genotype

Exploratory genotype-phenotype analyses were conducted within the genotyped cohort, including the Type 1 diabetes descriptive group, using participants with the data required for each outcome. Median C-peptide concentrations were 0.79 (0.70, 0.86) nmol/L in GT carriers and 0.70 (0.50, 0.83) nmol/L in GG carriers ($P = 0.157$). Median HOMA-IR was 3.0 (1.9, 3.7) in GT carriers and 2.7 (1.6, 3.6) in GG carriers ($P = 0.609$) (Table 3). Because only six GT carriers were identified

and the Type 1 diabetes group was included only for descriptive comparison, these exploratory findings should be interpreted cautiously.

Multivariable analysis

In the adjusted logistic regression model, rs12255372 GT carrier status was not independently associated with Type 2 diabetes (adjusted OR 0.646, 95% CI 0.070, 5.969; P = 0.700). Age was positively associated with Type 2 diabetes case status (adjusted OR 1.323, 95% CI 1.199, 1.459; P < 0.001), BMI showed an inverse association (adjusted OR 0.752, 95% CI 0.677, 0.835; P < 0.001), and men had higher odds of case status than women (adjusted OR 3.077, 95% CI 1.020, 9.283; P = 0.046) (Table 4).

Discussion

The principal finding of this study was the very low frequency of the rs12255372 T allele in the examined Indonesian cohort. Only six heterozygous participants were identified, no TT homozygotes were observed, and no detectable association with Type 2 diabetes was found. C-peptide and HOMA-IR also did not differ significantly by genotype. These findings should not be interpreted as evidence that TCF7L2 has no role in Type 2 diabetes; rather, they indicate that this particular marker had limited discriminatory value in a sample with very few minor-allele carriers.

The allele distribution differs from reports in several European, North African, Middle Eastern and South Asian populations, where the T allele is more frequent and has more often been associated with Type 2 diabetes.[9-15] This contrast supports the population dependence of genetic susceptibility signals. The cohort included participants from Banjar, Dayak, Javanese, Bugis, Butonese and Minangkabau backgrounds, illustrating substantial population diversity within Indonesia. Differences in allele frequency, linkage disequilibrium and haplotype structure can alter the measurable effect of a single marker. Accordingly, a risk variant identified in one population should not automatically be assumed to have the same predictive value in another.

TCF7L2 remains biologically relevant to pancreatic beta-cell function, incretin signalling, insulin secretion and glucose homeostasis.[5,8,16-18] The concurrent assessment of C-peptide and HOMA-IR was intended to explore whether rs12255372 tracked with endogenous insulin secretion or insulin resistance. The absence of statistically significant differences must be interpreted in the context of

only six GT carriers. The wide confidence interval around the Type 2 diabetes association likewise indicates substantial uncertainty rather than evidence of equivalence.

33 The adjusted model did not identify genotype as an independent correlate of Type 2 diabetes. Age and sex were positively associated with case status, whereas BMI showed an inverse association. Because cases and controls were not matched for age or sex, residual confounding is possible despite multivariable adjustment. These clinical associations therefore should not be interpreted causally and were included primarily to account for group imbalance in the genetic model.

Strengths of the study include Sanger sequencing-based genotype classification and concurrent evaluation of biochemical measures of glucose metabolism. The study also adds data from Indonesia, a population that remains under-represented in global diabetes genetics. Recent research increasingly emphasises ancestry-aware and multi-population approaches to risk architecture and functional interpretation of non-coding variants.[21-27] The present findings reinforce the need to validate candidate genetic markers locally before incorporating them into risk prediction or stratification.

Several limitations should be considered. The minor allele was rare, no TT homozygotes were identified and the number of GT carriers was insufficient for precise genotype-stratified estimates. Only one TCF7L2 variant was examined, preventing haplotype or multi-variant analysis. Recruitment was limited to a single Indonesian region and the modest sample size limits generalisability across the country. The exploratory genotype-phenotype analyses also included the small Type 1 diabetes descriptive group, which may influence C-peptide and HOMA-IR comparisons. Finally, the study did not include ancestry-informative markers, so the observed population differences cannot be attributed to genetic ancestry directly.

In conclusion, rs12255372 was uncommon in this Indonesian cohort and showed no detectable association with Type 2 diabetes, C-peptide or HOMA-IR. The findings highlight the importance of population-specific validation of diabetes-associated variants and support larger multicentre studies incorporating broader Indonesian representation, multiple TCF7L2 variants and ancestry-informed analyses.

Authors' contributions

PS contributed to conceptualisation, methodology, investigation, data curation, formal analysis and preparation of the original draft. MF contributed to bioinformatics-related analysis, interpretation of molecular findings and critical revision of the manuscript. HS contributed to clinical methodology, interpretation of clinical findings and critical revision of the manuscript. AAZ contributed to formal analysis, data interpretation and critical revision of the manuscript. RBL contributed to laboratory investigation, PCR amplification, Sanger sequencing validation and critical revision of the manuscript. MA and LBK contributed to conceptualisation, methodology, supervision and critical revision of the manuscript. PS and LBK confirm the authenticity of all raw data. All authors read and approved the final manuscript.

Data availability statement

The data generated in this study are available from the corresponding author on reasonable request.

References

1. Dilworth L, Facey A, Omoruyi F. Diabetes mellitus and its metabolic complications: the role of adipose tissues. Int J Mol Sci. 2021;22:7644.
2. Galicia-Garcia U, Benito-Vicente A, Jebari S, Larrea-Sebal A, Siddiqi H, Uribe KB, Ostolaza H, Martin C. Pathophysiology of type 2 diabetes mellitus. Int J Mol Sci. 2020;21:6275.
3. Genitsaridi I, Salpea P, Salim A, Sajjadi SF, Tomic D, James S, Thirunavukkarasu S, Issaka A, Chen L, Basit A, et al. 11th edition of the IDF Diabetes Atlas: global, regional, and national diabetes prevalence estimates for 2024 and projections for 2050. Lancet Diabetes Endocrinol. 2026;14:149-156.
4. Singh S, Kriti M, Anamika KS, Sarma DK, Verma V, Nagpal R, Mohania D, Tiwari R, Kumar M. Deciphering the complex interplay of risk factors in type 2 diabetes mellitus: a comprehensive review. Metab Open. 2024;22:100287.
5. Del Bosque-Plata L, Martinez-Martinez E, Espinoza-Camacho MA, Gragnoli C. The role of TCF7L2 in type 2 diabetes. Diabetes. 2021;70:1220-1228.
6. Huang ZQ, Liao YQ, Huang RZ, Chen JP, Sun HL. Possible role of TCF7L2 in the pathogenesis of type 2 diabetes mellitus. Biotechnol Biotechnol Equip. 2018;32:830-834.
7. Witka BZ, Oktaviani DJ, Marcellino M, Barliana MI, Abdulah R. Type 2 diabetes-associated genetic polymorphisms as potential disease predictors. Diabetes Metab Syndr Obes. 2019;12:2689-2706.
8. Zeini M, Laurenti MC, Egan AM, Muthusamy K, Ramar A, Vella E, Bailey KR, Cobelli C, Dalla Man C, Vella A. The longitudinal effect of diabetes-associated variation in TCF7L2 on islet function in humans. Diabetes. 2024;73:1440-1446.
9. Elhourch S, Arrouchi H, Mekkaoui N, Allou Y, Ghrifi F, Allam L, Elhafidi N, Belyamani L, Ibrahimi A, Elomri N, et al. Significant association of polymorphisms in the TCF7L2 gene with a higher risk of type 2 diabetes in a Moroccan population. J Pers Med. 2021;11:461.
10. Kaur N, Kaur G, Kaur S. TCF7L2 gene rs12255372 (G/T) variant and susceptibility to type 2 diabetes mellitus in North Indians. Gene Rep. 2020;19:100595.
11. Mahmood SA, Uddin MF, Mustari M, Mahjabeen S, Azad MK, Rahim CFM. TCF7L2 rs12255372 gene polymorphism and obesity in a Bangladeshi population. Indian J Endocrinol Metab. 2025;29:313-318.

12. Ghosheh N, Naffa RG, Mashal S, Al-Khalayfa S, Battah AH, Azab B. The association of TCF7L2 gene polymorphisms, rs12255372 and rs7903146, with type 2 diabetes mellitus in the Jordanian population. Mol Biol Rep. 2025;52:461.

13. Lu YC, Yu TH, Hsuan CF, Hsu CC, Hung WC, Wang CP, Tang WH, Cheng MC, Chung FM, Lee YJ, et al. Association of TCF7L2 rs7903146 (C/T) polymorphism with type 2 diabetes mellitus in a Chinese population: clinical characteristics and ethnic context. Diagnostics (Basel). 2025;15:2110.

14. Sen P, Alam SS, Jahan I, Saha P, Shill LC. Association of the TCF7L2 rs7903146 variant with type 2 diabetes mellitus and related biomarkers: a systematic review. BMJ Nutr Prev Health. 2025;8:e001205.

15. Blanken CPS, Bayer S, Buchner Carro S, Hauner H, Holzapfel C. Associations between TCF7L2, PPARG, and KCNJ11 genotypes and insulin response to an oral glucose tolerance test: a systematic review. Mol Nutr Food Res. 2025;69:e202400561.

16. Evans-Molina C. Advances in basic biology and translation in islet research. Nat Rev Endocrinol. 2025;21:69-70.

17. Adeerjiang Y, Sidike A, Gan XX, Li QT, Jiang S. The role of the Wnt3a/beta-catenin/TCF7L2 pathway in diabetes and cardiorenal complications. Cardiorenal Med. 2025;15:72-82.

18. Ansari S, Khoo B, Tan T. Targeting the incretin system in obesity and type 2 diabetes mellitus. Nat Rev Endocrinol. 2024;20:447-459.

19. American Diabetes Association Professional Practice Committee. 2. Diagnosis and classification of diabetes: Standards of Care in Diabetes-2025. Diabetes Care. 2025;48:S27-S49.

20. Little J, Higgins JPT, Ioannidis JPA, Moher D, Gagnon F, von Elm E, Khoury MJ, Cohen B, Davey-Smith G, Grimshaw J, et al. Strengthening the Reporting of Genetic Association Studies (STREGA): an extension of the STROBE statement. PLoS Med. 2009;6:e1000022.

21. Ortega HI, Udler MS, Gloyn AL, Sharp SA. Diabetes mellitus polygenic risk scores: heterogeneity and clinical translation. Nat Rev Endocrinol. 2025;21:530-545.

22. Plengvidhya N, Teerawattanapong N, Narkdontr T, Innang S, Songlilitchuwong S, Suthon S, Tangjittipokin W. Cluster-specific genetic associations of CDKAL1, CDKN2A, CDKN2B, HHEX, KCNQ1, MTNR1B, PAX4, SLC30A8, TCF7L2, and UBE2E2 variants in new onset type 2 diabetes. Sci Rep. 2026;16:2983.

23. Maynard AG, Bhardwaj R, Jones TR, Claussnitzer M. Bridging the variant-to-function gap in type 2 diabetes: advances and challenges. Diabetologia. 2026;69:274-294.

24. Hansen AL, Andersen MK, Engelhard LM, Brøns C, Hansen T, Nielsen JS, Vestergaard P, Højlund K, Jessen N, Olsen MH, et al. Impact of TCF7L2 rs7903146 on clinical presentation and risk of complications in patients with type 2 diabetes. Diabetes Obes Metab. 2025;27:2002-2011.

25. Yau B, Ghislain J, Kebede MA, Hughes J, Poitout V. The role of the beta cell in type 2 diabetes: new findings from the last 5 years. Diabetologia. 2025;68:2092-2103.

26. Mahajan A, Taliun D, Thurner M, Robertson NR, Torres JM, Rayner NW, Payne AJ, Steinthorsdottir V, Scott RA, Grarup N, et al. Fine-mapping type 2 diabetes loci to single-variant resolution using high-density imputation and islet-specific epigenome maps. Nat Genet. 2018;50:1505-1513.

27. Suzuki K, Hatzikotoulas K, Southam L, Taylor HJ, Yin X, Lorenz KM, Mandla R, Huerta-Chagoya A, Melloni GEM, Kanoni S, et al. Genetic drivers of heterogeneity in type 2 diabetes pathophysiology. Nature. 2024;627:347-357.

Table 1. Clinical and biochemical characteristics of study participants

Variable	Type 2 diabetes (n=111)	Controls without diabetes (n=60)	Type 1 diabetes (n=10)	P value
Age (years)	40 ± 7	32 ± 7	25 ± 6	<0.001
Sex (men/women)	23/88	4/56	2/8	0.001
BMI (kg/m²)	23.2 ± 1.4	23.6 ± 1.7	22.7 ± 1.8	0.008
Fasting glucose (mmol/L)	8.82 (7.88, 10.71)	4.83 (4.55, 5.05)	11.04 (10.55, 11.66)	<0.001
HbA1c, mmol/mol (%)	73 ± 18 (8.8 ± 1.7)	32 ± 2 (5.1 ± 0.2)	68 ± 13 (8.4 ± 1.1)	<0.001
Insulin (mIU/L)	9.8 ± 10.4	7.4 ± 1.4	2.0 ± 0.3	<0.001
C-peptide (nmol/L)	0.72 (0.53, 0.84)	0.77 (0.55, 0.84)	0.10 (0.07, 0.10)	<0.001
HOMA-IR	3.5 (2.8, 3.9)	1.5 (1.3, 1.7)	1.0 (0.9, 1.1)	<0.001

Data are mean ± standard deviation, median (interquartile range), or counts. Column headings show enrolled group sizes; summaries use available data for each variable. P values were calculated using one-way analysis of variance for normally distributed continuous variables, the Kruskal-Wallis test for non-normally distributed continuous variables, and Fisher's exact test for sex. BMI, body mass index; HbA1c, glycated haemoglobin; HOMA-IR, homeostasis model assessment of insulin resistance.

Table 2. Distribution of rs12255372 genotypes and alleles and association with Type 2 diabetes

Measure	Type 2 diabetes (n=111)	Controls without diabetes (n=60)	OR (95% CI)	P value
GG genotype	107 (96%)	58 (97%)	Reference	—
GT genotype	4 (3.6%)	2 (3.3%)	1.08 (0.19, 6.16)	0.930
G allele	218 (98%)	118 (98%)	—	0.920
T allele	4 (1.8%)	2 (1.7%)	—	—

Data are n (%). P values were calculated using Fisher's exact test. The genotype distribution in the control group was consistent with Hardy-Weinberg equilibrium (P = 1.000). CI, confidence interval; OR, odds ratio.

Table 3. C-peptide and HOMA-IR according to rs12255372 genotype

Outcome	GG (n=175)	GT (n=6)	P value
C-peptide (nmol/L)	0.70 (0.50, 0.83)	0.79 (0.70, 0.86)	0.157
HOMA-IR	2.7 (1.6, 3.6)	3.0 (1.9, 3.7)	0.609

Data are median (interquartile range). Column headings show genotype group sizes. Exploratory analyses used available data for each outcome within the genotyped cohort, including the Type 1 diabetes descriptive group. P values were calculated using the Mann-Whitney U test. HOMA-IR, homeostasis model assessment of insulin resistance.

Table 4. Multivariable logistic regression model for Type 2 diabetes

Variable	Adjusted OR	95% CI	P value
rs12255372 genotype	0.646	0.070, 5.969	0.700
Age	1.323	1.199, 1.459	<0.001
BMI	0.752	0.677, 0.835	<0.001
Sex	3.077	1.020, 9.283	0.046

Adjusted ORs were estimated using multivariable logistic regression for the primary case-control analysis, including genotype, age, BMI and sex in the model. Outcome coding: controls=0 and Type 2 diabetes=1. Genotype coding: GG=0 and GT=1. Sex coding: women=0 and men=1. BMI, body mass index; CI, confidence interval; OR, odds ratio.

Figure legend

Figure 1. Representative Sanger sequencing chromatograms of TCF7L2 rs12255372. (a) GG genotype, showing a single guanine peak at the polymorphic locus. (b) GT genotype, showing overlapping guanine and thymine peaks.